AL SAFA 2 PARK

Dubai Municipality — AI-Powered Park Design Challenge

DELIVERABLE 10

AI Methodology Report

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Maintained as a living document throughout the project and compiled here in full.

10.1 Purpose and Toolchain

This project's competitive premise is that AI/data methodology — not architectural training — can produce a more defensible, better-grounded park design. This report documents how that actually happened: which tool did what, what each AI-driven decision was based on, and every point where an AI output was checked, corrected, or rejected rather than trusted.

Tool	Role
Claude (this project)	Strategy, narrative, methodology documentation, spatial/data reasoning, independent verification, and compiling every submission-ready deliverable
Claude Code	File parsing (DWG/DXF), geometric computation, OSM/Overpass queries, GIS analysis, spatial layout generation
Gemini Pro / Nano Banana	All visual design outputs — concept renders, masterplan visualization, materials mood-boarding
GPT Pro	Critique/proofreading pass only — not used for primary drafting

10.2 Methodology Principles

Three principles were applied consistently from the first working session to the last, not adopted partway through:

- **Three-tier confidence tagging —** every claim across every deliverable is marked [CONFIRMED] (verified against source data), [INFERRED] (reasoned from confirmed data, not independently verified), or [GAP] (explicitly missing, not silently filled). This is the single most consistent methodological thread across the whole project.
- **Verify before compiling, not after submitting —** every Claude Code output was independently re-derived from raw geometry/data before being written into a deliverable, not accepted because it was well-formatted or confidently stated.
- **Surface conflicts, don't silently resolve them —** both AI tools were explicitly instructed to flag competing requirements, ambiguous data, and standards that don't apply cleanly, rather than picking a plausible answer and moving on.

10.3 Chronological Log

Stage 1 — Site Analysis: Geometry

[2026-07-26] Attempted local DWG parsing — blocked by missing tooling/network access, correctly routed to Claude Code rather than approximated. Claude Code converted the DWG via ODA File Converter and found the source file contained three duplicate, ambiguous plot tracings from a PDF-import artifact, with no single obviously-correct boundary. Rather than guess, Claude Code surfaced full data on all three candidates.

Claude independently resolved the ambiguity: pixel-coordinate analysis of the generated plot identified an unclassified traced circle matching the same walkability-catchment device used in Dubai Municipality's own locator graphic, geometrically centered on one specific candidate boundary — resolving which of three tracings was authoritative without simply picking one.

Claude Code then isolated that region and extracted boundary vertices, tree points, and hatch geometries. Claude independently verified the boundary via shoelace area, side-length, and interior-angle calculation from raw coordinates — confirming a 163×91m rectangle (89.3°–91.1° corners) matching the drawing's own dimension text, upgrading the boundary from working hypothesis to geometrically confirmed.

Stage 1 — Site Analysis: Real-World Verification

Claude cross-validated the DWG's embedded coordinate text against the real-world park location via independent UTM Zone 40N conversion — a 34m northing match against Google Places geolocation, upgrading confidence from "geometrically self-consistent" to "verified against ground truth." This also surfaced live park amenities and a real user-reported accessibility/equity issue (court-access gatekeeping) not present in any official document, and sourced Dubai climate design parameters (temperature, humidity, prevailing wind) from current data rather than generic assumptions.

Stage 1 — Site Analysis: Orientation and Solar Geometry

Claude Code queried OpenStreetMap directly for the tagged park polygon rather than inferring from nearby streets, cross-matched all 4 edges by length against the CAD boundary (agreement within 1.5°), and found the DXF required negligible rotation correction (~0.5°) to align with true north. Claude independently re-derived the bearing conversion from raw vertex coordinates to catch potential sign/quadrant errors before trusting the result, then computed solar altitude/azimuth for summer/winter solstice at the site's exact latitude — converting an abstract orientation fact into a concrete, edge-by-edge environmental prioritization used throughout every later deliverable.

Stage 1 — User & Community Analysis

Personas and accessibility strategy were built directly from confirmed site adjacencies (school, rehab center, mosque, shopping center) and real review evidence, rather than generic park-design assumptions. Claude sourced the applicable accessibility standard (Dubai Universal Design Code, issued by the client itself) rather than assuming a generic international code — surfacing the adjacent rehab center as a daily accessibility user, not a hypothetical one.

Stage 2 — Concept & Spatial Organization

Claude Code generated the first concept spatial layout; Claude independently recomputed the site's land-use breakdown and found the source report's own setback-strip figure was mislabeled (~1,139 m² stated vs. 1,488.6 m² true geometric figure, verified by direct polygon inset calculation) — a real arithmetic inconsistency, not a rounding difference, caught before it reached a deliverable.

Precise edge-to-neighbor mapping (mosque, rehab center, school) was resolved via true perpendicular point-to-edge distance against the real OSM polygon, cross-checked with Google Places for facilities OSM couldn't identify by tag. "Shopping Center" was left explicitly unresolved after two independent searches rather than forced onto the nearest plausible building — a deliberate choice to preserve an honest gap over false confidence.

Stage 3 — Preliminary Masterplan: A Real Bug, Found and Fixed

Claude Code's masterplan revision introduced a genuine software bug: three path spurs (Playground, Picnic Grove, Restroom) were computed for length but had their geometry discarded before export, while a separate quantities file was double-counting one spur. This was caught not by inspecting the geometry directly but by noticing the same "loop + spurs" figure was reported as three different numbers across three of Claude Code's own output files — an internal-consistency check, not a geometric one. Claude Code traced the exact root cause, rebuilt the spur logic around a single shared source of truth, and Claude independently re-verified the fix at the geometry level: all 7 spurs confirmed as real LineString features, each with one endpoint exactly on the path loop and the other exactly on its destination's boundary (0.00m in both cases) — not just a corrected number, but confirmed real connectivity.

Stage 4 — Sections & Elevations

Claude Code correctly deferred to confirmed Z=0 CAD ground truth over a coarse elevation-API artifact (a clean -2.0m/3.0m step pattern recognized as a ~30m-cell DEM grid boundary, not real topography) rather than assuming either flatness or unverified grade changes. Claude independently re-verified every cited height standard against current sources: FIBA 3x3's "no enclosure standard" held up exactly; the cited Padel side-wall height (~1.0m) did not — corrected to ~3.0m against multiple independent FIP-derived sources. A specific,

checkable claim about a real sports governing body's specification was wrong and was caught before submission.

Stage 5 — Materials, Landscape, and a Cross-Deliverable Compliance Catch

Sourcing real Dubai Municipality landscape practice (for Deliverable 7) surfaced an unrelated but material compliance gap in the already-confirmed Deliverable 4 masterplan: Dubai Municipality requires 70-80% shade coverage over active play areas, and the existing design provided 0%. This was found only because materials research for one deliverable was checked against the geometry of an earlier, already-"complete" deliverable, rather than treating each deliverable as sealed once compiled.

Claude Code confirmed the 0% gap geometrically (not assumed), proposed two shade canopies sized exactly to the requirement, and flagged real construction/accounting caveats unprompted — that the proposed spans exceed practical single-sail-module limits, and that the sail area must not be double-counted against ground already classified as hard surface. After approval, Claude independently re-verified the built fix via direct polygon clipping against the actual delivered geometry (76.5%/73.7% coverage, exact match) rather than trusting the reported percentages.

Separately, an incorrect standard citation (CPSC, a US standard) was identified and corrected to EN 1176/1177 — the standard actually recognized by Dubai Municipality, Trakhees, and KHDA — and swept across every affected deliverable rather than only the one being actively edited.

Stage 6 — Visualization

A written grounding brief (real species, real materials, real dimensions, explicit exclusions) was compiled before any image generation, specifically to prevent generic "desert park" output. Every returned render was checked against the confirmed masterplan and materials brief before compiling Deliverable 6: most held up, but the first site-overview render was flagged for two real drift issues — garbled legend text and rounded boundary corners contradicting the geometrically-verified rectangular boundary. The regenerated version was re-checked and confirmed fixed, with one minor labeling note carried forward rather than triggering a further regeneration cycle.

Stage 7 — Sustainability and UX Synthesis

Deliverable 9 scoped Dubai's Al Sa'fat green building standard precisely — it is building-envelope focused and applies only to the design's one permitted structure (the restroom), not the landscape as a whole — rather than citing it as a blanket "we comply with green standards" claim. Deliverable 8 synthesized existing evidence (the confirmed court-access equity complaint, the real on-site cafeteria, the confirmed climate/activation patterns) into a concrete operational recommendation, rather than introducing new unverified assumptions about user behavior.

Stage 8 — Scenario Testing: Two Alternatives, Scored Against Confirmed Data

Two alternative spatial concepts ("Central Spine" — a conventional symmetric axial layout; "Concentric Core" — activity radiating from a central hub) were generated from the same confirmed site geometry and scored against the site's actual verified constraints: edge-specific solar/wind exposure, the 28.7m mosque-buffer projection point, the real 74-tree distribution, and confirmed neighbor adjacency. Both alternatives are legitimate, commonly-used park-design patterns — the comparison was not built to make them look artificially weak. Each failed on identifiable, data-grounded points: both placed active courts against the mosque-facing edge with no buffer; Central Spine ignored that the NW edge is the site's harshest exposure zone; Concentric Core's circular geometry fit poorly inside the rectangular site. One genuine strength surfaced from Alternative B — its naturally continuous perimeter circulation loop — was explicitly carried forward into the final concept's accessible path design rather than discarded along with the rest of that alternative.

Full comparison methodology, scoring matrix, and diagrams: Deliverable 2.

10.4 Notable AI-Driven Corrections — Summary

Every genuine error, ambiguity, or gap caught during this project, in one place:

Issue	Caught by	Method
Ambiguous DWG boundary (3 duplicate tracings)	Claude	Pixel-coordinate analysis matching a traced catchment circle to the official locator graphic
Setback-strip area mislabeled in land-use table	Claude	Independent polygon-inset area recalculation
3 path spurs missing from GeoJSON export + duplicate spur count	Claude Code (diagnosis/fix) + Claude (re-verification)	Cross-file consistency check; geometric re-derivation of connectivity
Padel court side-wall height cited incorrectly (~1.0m vs ~3.0m)	Claude	Independent web verification against multiple FIP-derived sources
0% shade coverage over both playgrounds vs. 70-80% DM requirement	Claude (surfaced via unrelated research) + Claude Code (fix) + Claude (re-verification)	Cross-deliverable check; direct polygon-clipping verification of the fix
Wrong regional playground standard cited (CPSC vs. EN 1176/1177)	Claude	Verification against DM/Trakhees/KHDA-recognized standards
Rendered site-overview boundary geometry (rounded vs. rectangular)	Claude	Visual QA against confirmed geometric data

10.4a AI Application Crosswalk — Against the Brief's Own Categories

The Scope of Work explicitly lists six categories of AI application it encourages. Mapped directly against what was actually done in this project, with no category left unaddressed:

Brief's AI application category	What this project actually did
Site Analysis and Opportunities Assessment	CAD geometry resolution via pixel-coordinate analysis; UTM geodetic cross-validation against real-world location; OSM boundary/edge-bearing confirmation (Deliverable 1)
User and Community Analysis	Personas and accessibility strategy built from confirmed adjacency data and live review evidence, not generic assumption; Dubai Universal Design Code sourced directly (Deliverable 1)
Environmental Analysis (solar, wind, thermal, microclimate)	Solar altitude/azimuth computed at the site's exact confirmed latitude for both solstices; edge-specific exposure prioritization; independent DEM-artifact detection for ground-truth flatness (Deliverables 1, 5)
Design Ideation and Concept Generation	Two alternative spatial concepts generated and diagrammed from the same confirmed geometry (Central Spine, Concentric Core) alongside the chosen Threshold Park concept (Deliverable 2)
Design Development and Optimization	Scored, criteria-based comparison of all three concepts against confirmed site constraints; iterative masterplan revision cycle (Deliverables 3-4) including a real compliance-gap fix verified via direct polygon geometry (Deliverable 4, Section 4.6a)
Sustainability and Resilience	Climate-responsive zoning built into the concept from the start, not applied retroactively; recycled-water irrigation strategy; Al Sa'fat standard scoped precisely to what it actually governs (Deliverable 9)

This crosswalk exists so the connection between the brief's stated intent and this project's actual work is explicit and checkable, rather than left for a reviewer to infer.

- **True-north orientation** is accurate to approximately ±1.5° (OSM crowd-tracing tolerance), not survey precision — adequate for concept-stage design, flagged everywhere it's used.
- **Tree conflicts (12 of 74)** are point-based, not canopy/trunk-based — the true number requiring removal could be smaller with a real arborist survey.
- **"Shopping Center" neighbor** was never identified by name or exact edge — left as a standing gap rather than a forced guess, across two independent search methods.
- **No specific irrigation water budget, solar PV feasibility study, or NW-edge vehicle easement** has been calculated or confirmed — each flagged in its relevant deliverable as a next step for the appropriate specialist, not silently assumed.
- **Corner-overlap simplifications** in the schematic land-use geometry are acceptable for a concept diagram and explicitly flagged as unsuitable for direct use in construction CAD.

10.6 Reflection

The recurring pattern across this entire project was not "AI got it right" or "AI got it wrong" — it was that useful design decisions came from treating every AI output, including this project's own, as a claim to verify rather than a fact to accept. The clearest value wasn't generating content faster; it was that cross-checking outputs against each other and against raw geometry surfaced real errors — a missing path connection, a mislabeled area, a wrong sports standard, a genuine regulatory compliance gap — that a single confident-sounding output, human or AI, would not have caught on its own.

This report compiles methodology notes maintained throughout the project (Deliverables 1-9) into a single document, per the brief's requirement that this log be maintained as a living document rather than reconstructed from memory at the end.